

## PROJECT AND TEAM INFORMATION

### Project Title

*Detective Mystery: The Hidden Truth*

### Student / Team Information

|                                          |                                                                                                                                                                            |
|------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Team Name :</b><br><b>Team ID :</b>   | <b>Truth Seekers</b><br><i>T152</i>                                                                                                                                        |
| <b>Team Member 1</b><br><i>Team Lead</i> | <b>Badhani, Karan</b><br>Student ID: 2510360037<br><i>karanbadhani9770@gmail.com</i><br>  |
| <b>Team Member 2</b>                     | <b>Giri Goswami, Suraj</b><br>Student ID: 2510012585<br><i>2510012585@geu.ac.in</i><br> |
| <b>Team Member 3</b>                     | <b>Singal, Khushi</b><br>Student ID: 2510360039<br><i>singalk20@gmail.com</i><br>       |
| <b>Team Member 4</b>                     | <b>Arora, Tanmay</b><br>Student ID: 2510610178<br><i>2510610178@geu.ac.in</i><br>       |

## PROPOSAL DESCRIPTION

### Motivation

*Learning Object-Oriented Programming (OOP) and Data Structures and Algorithms (DSA) is an essential part of computer science, but students often understand these concepts mainly through theoretical examples and isolated programming exercises. As a result, it can be difficult to understand how these concepts work together in a practical application.*

*Our project “The Hidden Truth” aims to address this gap through an interactive detective mystery game. In the game, the player takes the role of a detective investigating a fictional case by exploring locations, collecting clues, examining evidence, interacting with suspects and connecting pieces of information to identify the culprit. The investigation is designed so that the underlying game mechanics naturally use OOP and DSA concepts rather than simply presenting them as separate demonstrations.*

*Different entities such as the player, suspects, clues, evidence, locations and cases can be represented as objects, making concepts such as classes, encapsulation, inheritance and polymorphism meaningful within the application. Data Structures such as graphs, stacks, queues, hash-based structures, searching and sorting techniques can be used for navigation, investigation history, clue management, suspect analysis and efficient information retrieval.*

*The motivation behind the project is therefore to create an engaging mystery-solving experience and to demonstrate how fundamental programming concepts can be transformed into a practical, understandable and interactive software system.*

### State of the Art / Current Solution

*At present, OOP and DSA are commonly learned through lectures, coding exercises, laboratory programs and small algorithm-based problems. These methods are effective for learning individual concepts, but the connection between the concepts and a complete interactive application is not always clear to beginners. Students may know the definition of a stack, graph, hash table or inheritance yet still find it difficult to identify where these concepts would be useful in a real software system.*

*Games provide an opportunity to make this learning more interactive. Existing mystery and detective games generally focus on storytelling, exploration, clues and decision-making, while educational programming applications generally focus on algorithms and data structures separately.*

*Our proposed system combines these two approaches. Instead of teaching DSA as isolated theory, The Hidden Truth incorporates DSA and OOP into the actual investigation process. For example, a graph can represent the investigation map, a stack can maintain investigation history, hash-based structures can provide efficient clue retrieval, and searching/sorting can support evidence and suspect analysis. This creates a single application where programming concepts have a visible purpose within the gameplay.*

## Project Goals and Milestones

*The main goal of The Hidden Truth is to develop an interactive detective mystery game that demonstrates the practical integration of Object-Oriented Programming and Data Structures and Algorithms. The project aims to make the game engaging for the player while maintaining a clear and meaningful technical structure.*

### **Major Goals**

- 1. Design a fictional detective case involving suspects, locations, clues, evidence and a final deduction.*
- 2. Apply OOP principles such as classes and objects, encapsulation, inheritance, polymorphism and abstraction.*
- 3. Apply appropriate DSA concepts such as graphs, stacks, queues, hash-based structures, searching and sorting.*
- 4. Develop an investigation system in which the player's actions and collected information influence the progress of the case.*
- 5. Provide efficient searching and organization of clues, evidence and suspect information.*
- 6. Develop a functional prototype that can be demonstrated and evaluated.*

### **Initial Milestones**

**Milestone 1:** *Finalize the game story, case structure, characters, locations and investigation flow.*

**Milestone 2:** *Identify the OOP classes and relationships and prepare the basic system architecture.*

**Milestone 3:** *Map each gameplay feature to an appropriate data structure or algorithm.*

**Milestone 4:** *Develop the core player, location, clue, suspect and evidence modules.*

**Milestone 5:** *Implement investigation navigation, clue collection, searching and evidence management.*

**Milestone 6:** *Integrate the major DSA and OOP components and test their interaction.*

**Milestone 7:** *Complete the playable prototype, perform testing, fix logical errors and prepare the final demonstration.*

*The final milestone is to demonstrate a complete investigation in which the player can gather information, analyze it and reach a logical conclusion.*

## Project Approach

*The project will be developed as a modular detective mystery application using C++, with Object-Oriented Programming forming the main software design approach and appropriate Data Structures and Algorithms supporting the investigation mechanics. The prototype will initially focus on functionality and logical gameplay, with graphical or interface improvements added according to the available development time.*

*The game will begin with a fictional case and an investigation map containing multiple locations. The player will be able to move between locations, interact with suspects, discover clues, collect evidence and review previously obtained information. Each major entity will be designed as a class. Possible classes include Player, Case, Suspect, Clue, Evidence, Location, InvestigationManager and GameManager. This structure will allow encapsulation of related data and functions and provide opportunities to demonstrate inheritance and polymorphism where different types of entities require different behaviours.*

*DSA will be integrated according to the actual requirements of the game. A graph will represent locations and their connections, allowing the investigation to be treated as a navigation problem. A stack can maintain the player's investigation history and support backtracking. A queue will maintain pending investigation tasks in the order they are generated. A hash-based structure can provide efficient retrieval of clues or evidence using unique identifiers. Searching algorithms will help locate relevant information, while sorting can rank suspects or evidence according to selected criteria.*

*Development will proceed incrementally: story and requirements, system design, class structure, data-structure mapping, core implementation, integration, testing, debugging and final prototype demonstration. This approach ensures that every technical concept has a clear purpose within the gameplay.*

## System Architecture (High-Level Diagram)

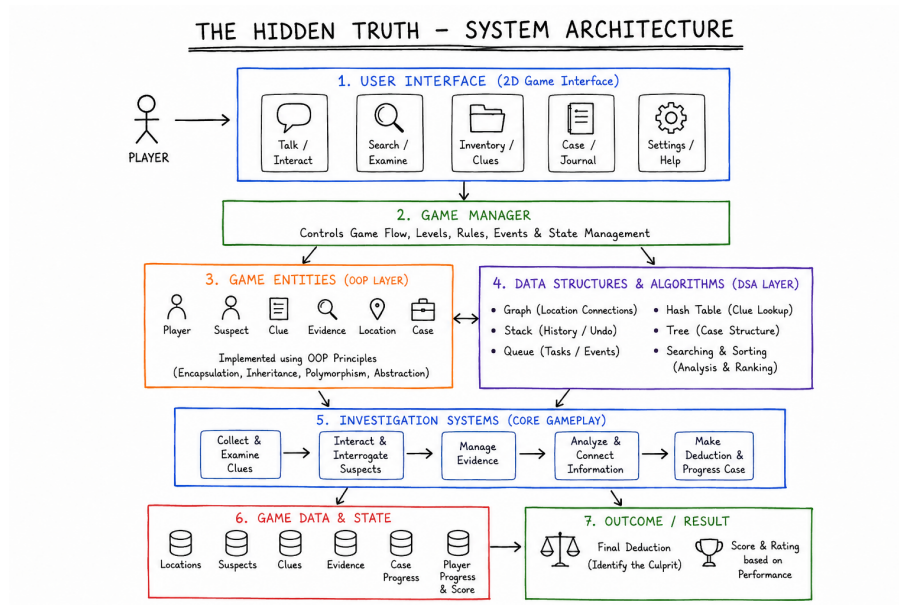


Figure 1: High-level system architecture of The Hidden Truth

## Project Outcome / Deliverables

The expected outcome of the project is a functional prototype of The Hidden Truth – an interactive detective mystery game demonstrating the practical use of OOP and DSA concepts.

The major deliverables will include:

- A complete fictional detective case with a structured investigation storyline.
- A playable investigation environment containing multiple locations, suspects, clues and evidence.
- An OOP-based software structure containing classes and relationships for major game entities.
- Implementation of selected data structures and algorithms for navigation, clue management, investigation history, information retrieval and evidence/suspect analysis.
- A working deduction mechanism through which the player uses collected information to identify the most likely culprit.
- A high-level system architecture and documentation explaining the relationship between gameplay features, OOP concepts and DSA concepts.
- A tested prototype suitable for demonstration during the project evaluation.

The final system is expected to demonstrate that fundamental programming concepts can be integrated into an application with a clear practical purpose rather than being implemented only as isolated academic examples.

## Assumptions

*The project assumes that the game will run on a standard computer with a suitable C++ development environment. The investigation will use a fictional case with original characters, locations, clues and evidence. The number of locations, suspects and clues will be limited initially to keep the prototype manageable within the project timeline. Players will interact through a keyboard-based or basic graphical interface. The project focuses on demonstrating OOP and DSA through gameplay and is intended as an academic prototype, not as a real-world criminal investigation system.*

## References

1. SFML Documentation: <https://www.sfml-dev.org/documentation/>
2. C++ Reference: <https://en.cppreference.com/>
3. GeeksforGeeks – Data Structures: <https://www.geeksforgeeks.org/data-structures/>
4. Microsoft Learn – C++ Documentation: <https://learn.microsoft.com/en-us/cpp/>
5. Programiz – Data Structures and Algorithms: <https://www.programiz.com/dsa>